C01 - ASSESSMENT TOOL

# Concept Mapping Tasks: Cloud Computing Fundamentals

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| **Title** | Concept Mapping Tasks - Cloud Computing |
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| **Course Code** | CSA1514 |
| **Course Name** | Cloud Computing and Big Data Analysis |

*Submitted as part of the Cloud Computing and Big Data Analysis coursework*

## Overview

Cloud computing has become the dominant paradigm for delivering computing resources, reshaping how individuals, businesses, and governments provision infrastructure, build applications, and store data. Rather than owning and maintaining physical hardware, organisations increasingly rent computing power, storage, and software as services delivered over the internet, paying only for what they consume. This shift has lowered the barrier to entry for startups, enabled rapid global scaling for established enterprises, and introduced entirely new operating models such as serverless computing and platform-based development.

This document presents four concept maps built around cloud computing fundamentals, each one designed to visually organise a distinct but interconnected aspect of the subject. Concept mapping is used deliberately here because it allows dense, interrelated technical material to be represented as a hierarchy of nodes connected by labelled, directional arrows (for example “provides”, “enables”, “includes”), making the relationships between ideas explicit rather than leaving them implicit in prose.

Together, the four maps progress logically from foundational definitions to increasingly technical detail. The first map establishes what cloud computing is, along with its defining characteristics, benefits, and limitations. The second map steps back historically, tracing how decades of parallel advances in hardware, software, and distributed systems converged to make modern cloud computing possible. The third map narrows the focus onto the specific service models offered by cloud providers today — IaaS, PaaS, SaaS, and XaaS — and clarifies the shared-responsibility boundary between provider and consumer for each. The fourth and final map drills into the two core technical enablers, virtualization and web services, that make elastic, on-demand cloud delivery technically achievable.

A consistent colour legend is used across all four maps so that a node's role can be recognised at a glance regardless of which map it appears in: dark navy ellipses mark the root or core concept of a map; rounded rectangles in blue, green, or purple denote major category or branch nodes; lighter shaded rectangles represent sub-concepts or detail nodes; yellow/gold nodes indicate outcomes or results; and red or pink nodes flag cross-cutting or governing concepts such as risks and the shared-responsibility model. Arrows throughout are labelled to state precisely how two connected nodes relate, which keeps each map readable as a standalone diagram while still allowing comparison across the whole set.

### Colour legend used across all four maps

- Root / Core Concept — dark navy ellipse marking the central topic of each map.
- Category / Branch Node — major sub-division of the root concept.
- Sub-concept / Detail Node — specific item nested under a branch node.
- Outcome / Result — a consequence or benefit flowing from another node.
- Cross-cutting / Governing Concept — an idea, risk, or rule that applies across branches.
- Arrows — labelled relationships describing exactly how two nodes are connected.

# Concept Map 1: Cloud Computing Fundamentals

This first map lays the conceptual groundwork for the rest of the document by defining cloud computing itself and then organising that definition into three branches radiating from the root node: the defining characteristics that distinguish cloud computing from traditional IT, the benefits that these characteristics make possible, and the limitations that organisations must plan for when adopting the cloud.

### Definition

Cloud computing is defined here as the on-demand delivery of computing resources — servers, storage, databases, networking, and software — over the internet on a pay-as-you-go basis. This definition deliberately mirrors the widely used NIST-style framing of cloud computing, emphasising two ideas at once: first, that resources are delivered remotely over a network rather than owned on-premises, and second, that consumption is metered and billed incrementally rather than paid for as a fixed upfront capital cost. The definition node sits directly beside the root node in the map, connected by an “is defined as” arrow, signalling that everything else in the map is a consequence of this core idea.

### Characteristics

The Characteristics branch expands into five essential-characteristic nodes that, together, are what make a service “cloud” rather than simply “remote”. On-Demand Self-Service means a consumer can provision computing capabilities automatically, without requiring human interaction with the provider. Broad Network Access means capabilities are available over the network through standard mechanisms, accessible from a wide range of client platforms. Resource Pooling refers to the provider's computing resources being pooled to serve multiple consumers using a multi-tenant model, with resources dynamically assigned according to demand. Rapid Elasticity describes the ability to scale capabilities elastically, sometimes automatically, to match workload in real time. Measured Service means resource usage is monitored, controlled, and reported, providing transparency for both provider and consumer under the pay-as-you-go model. In the diagram each of these five nodes is linked to the Characteristics branch node with an “includes” arrow, showing that they are not independent features but constituent parts of what “characteristics” means in this context.

### Benefits

The Benefits branch, connected to the root with a “provides” arrow, captures the practical advantages that organisations gain as a direct consequence of those characteristics. Cost Efficiency follows from not having to buy, house, and maintain physical hardware, converting capital expenditure into operating expenditure. Scalability follows from rapid elasticity, allowing capacity to grow or shrink with demand rather than being fixed by whatever hardware was originally purchased. Flexibility and Agility follow from on-demand self-service, letting teams provision new environments in minutes rather than weeks. Reliability and High Availability follow from resource pooling and provider-managed redundancy across data centres. Global Accessibility follows from broad network access, letting users and customers reach services from anywhere with an internet connection. Each of these five benefit nodes is linked back to the Benefits node with a “leads to” arrow, making explicit that they are outcomes rather than starting points.

### Limitations

The final branch, Limitations, is connected to the root with a “faces” arrow and represents the trade-offs and risks that come with adopting a cloud model, which organisations must weigh against the benefits above. Security and Privacy Risks arise because data and workloads now reside on infrastructure controlled by a third party. Internet Dependency means service availability is tied to network connectivity in a way that on-premises systems are not. Vendor Lock-in describes the difficulty of migrating away from a provider once an organisation is deeply integrated with its proprietary tools and APIs. Limited Control reflects the fact that the underlying infrastructure, and sometimes the platform or software itself, is managed by the provider rather than the consumer. Compliance and Legal Issues arise when data residency, industry regulation, or cross-border data transfer rules constrain where and how cloud resources can be used. Each of these five nodes is linked to the Limitations node with a “risk of” arrow, framing them consistently as risk factors rather than certainties.

Read as a whole, Concept Map 1 establishes a cause-and-effect narrative: a specific definition of cloud computing gives rise to five defining characteristics, which in turn produce a set of tangible benefits, while simultaneously exposing the organisation to a corresponding set of risks and limitations that must be actively managed. This framing sets up the more detailed maps that follow, which each expand on one part of this picture in greater depth.

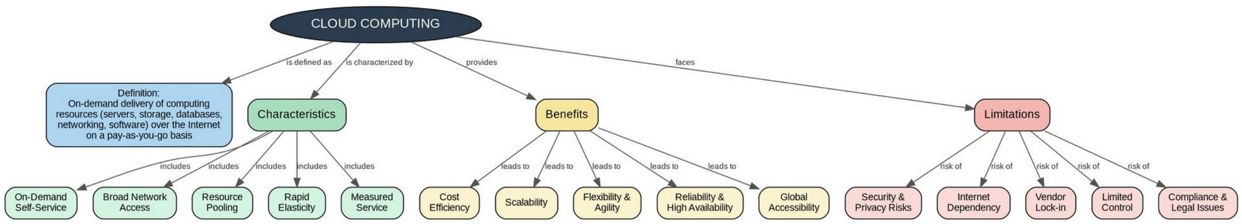


*Figure 1. Cloud Computing Fundamentals — definition, characteristics, benefits, and limitations.*

# Concept Map 2: Evolution of Cloud Computing

This map traces four converging historical threads — hardware evolution, internet/software evolution, distributed systems, and service delivery — and shows (via the dashed red arrows) how advances in one thread enabled progress in another, ultimately converging into modern cloud computing.

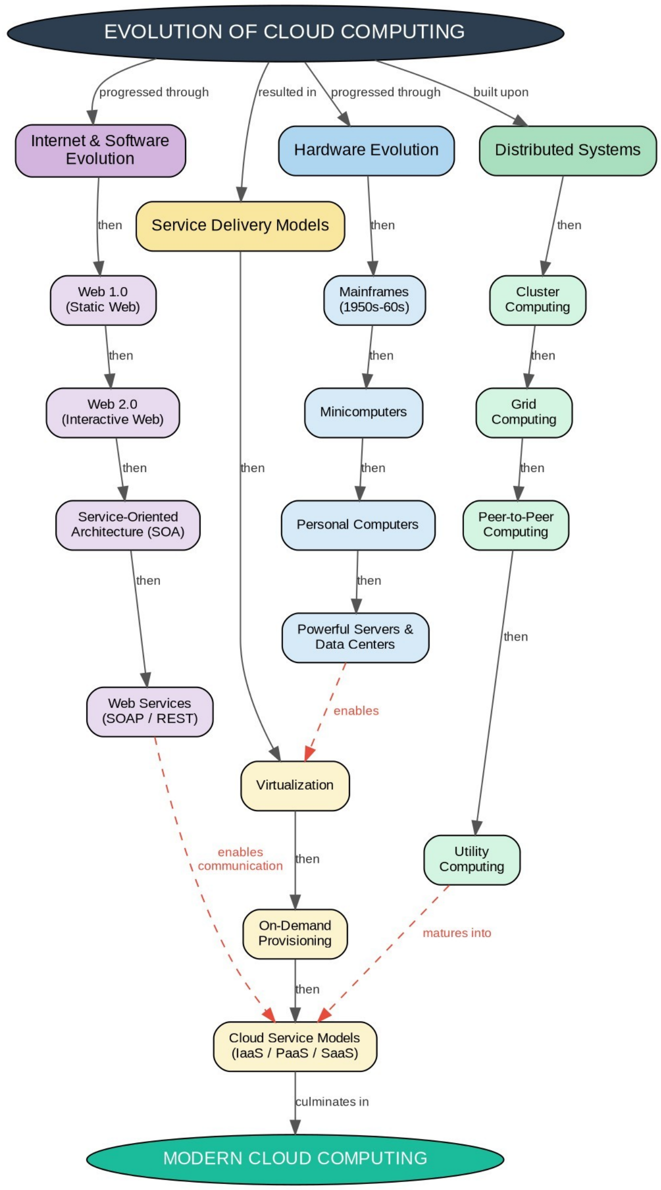


*Figure 2. Evolution of Cloud Computing — hardware, software, distributed systems, and service delivery.*

## Concept Map 3: Service Models

This map maps IaaS, PaaS, SaaS, and XaaS against the Shared Responsibility Model, showing exactly which layers of the stack the provider manages versus what remains the user's (consumer's) responsibility for each model, along with representative real-world examples.

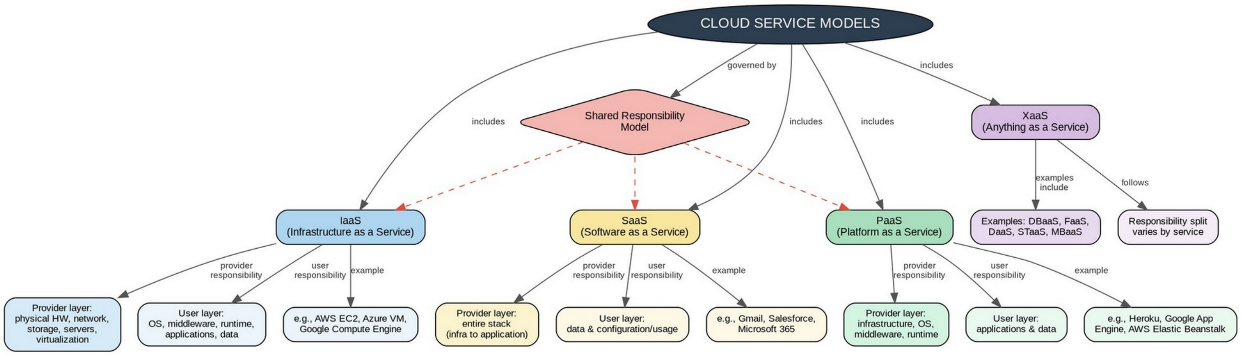


*Figure 3. Service Models — IaaS, PaaS, SaaS, and XaaS with provider vs. user responsibilities.*

# Concept Map 4: Virtualization and Web Services as Enablers of Cloud

This map shows how virtualization (hypervisor → virtual machines → hardware abstraction → multi-tenancy/resource pooling) and web services (API → SOAP/REST → interoperability) independently support scalability and service communication, and how both converge to enable on-demand cloud deployment and elastic scaling.

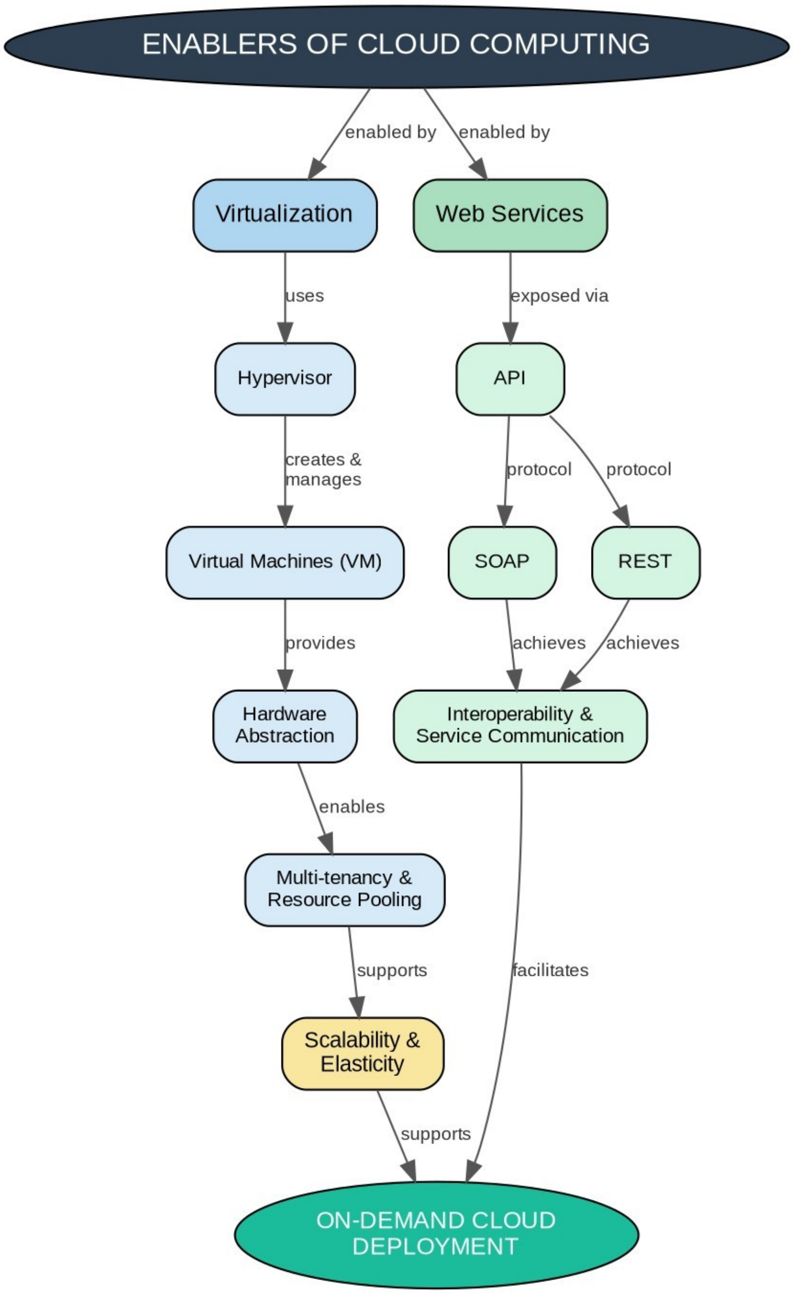


*Figure 4. Virtualization and Web Services as Enablers of Cloud — supporting deployment and scalability.*